SUPPORTING ADULT GUIDANCE

Using and checking formulae in context

Formulae and Rearranging · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This final lesson of the unit applies formulae to real problems and builds the habit of checking. Learners judge whether answers are believable and identify the likely slip, convert units before substituting and see the consequences of not doing so, work through decision trails (know, need, route, units, calculate, check), and plan a day out with several formulae and a budget.

Driving Question: *Does my answer make sense?*

The mathematics, in plain terms

Correct use of a formula requires consistent units, an appropriate route (direct substitution when the unknown is the subject; rearrangement or solving otherwise), and verification — by estimation, by reasonableness in context, and by substituting the result back into the original relationship.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Units first

A train travels 150 km in 90 minutes. Use s = d/t to find its speed in km per hour.

1. Know d = 150 km and t = 90 minutes; need s.	s is the subject: substitute
2. The answer is in km per hour, so the time must be in hours.	90 minutes = 1.5 hours
3. Substitute and calculate.	s = 150 ÷ 1.5 = 100
4. Check: sensible for a train? And 100 × 1.5 = 150 ✓	yes

100 km per hour.

Example 2 — Rearrange, then check

A taxi costs C = 3 + 2m pounds. How far can you go for £19?

1. Know C = 19; need m.	rearrange or substitute
2. Rearrange: m = (C – 3)/2.	m = (19 – 3)/2
3. Calculate.	m = 8
4. Check: 3 + 2 × 8 = 19 ✓	8 miles for £19 is sensible

8 miles.

Checking is part of the answer

Throughout the lesson, an answer is not complete until it has been checked. In the decision trail, a wrong unit choice shows where it would have led (for example, a train slower than walking). Encourage learners to say their sense check aloud: 'Is a 4 m sail believable?'

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The sense detector: believable or not, and what was the slip?	~6 min
Movement	Units first: choose the conversion, then calculate — and see what happens if you don't.	~8 min
Reflection	The decision trail: know, need, route, units, calculate, check.	~16 min
Creativity	Plan a day out with several formulae and a budget.	~10 min
Rest	Noticing formulae hidden in everyday price lists, recipes and timetables.	~2 min
Nutrition	Intellectual: judgement — checking before relying on an answer.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Substituting minutes into a formula that needs hours.
Ask: 'What units does the answer need? What units must the time be in?'

Mixing cm and m in an area.
Convert everything to the unit the answer needs before multiplying.

Multiplying distance by time to get speed.
Use a sense check: does riding for longer make you faster?

Accepting an answer without checking it.
Make the final check routine: substitute back, and picture it.

Thinking only one route is allowed.
Both substituting then solving and rearranging then substituting are correct; compare them.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: converting units of length, time and money, and Lessons 2 to 5 of this unit. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says a cyclist who rides 6 km in 30 minutes goes at 0.2 km per hour. The aim is a question that lets them find the slip themselves.

- "Is 0.2 km per hour faster or slower than walking?"
- "What units is your time in? What units does the speed need?"

- “How many hours is 30 minutes?”
- “If you ride at your speed for 30 minutes, how far do you go?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Do answers need units?
In the Practice Companion, numbers alone are accepted, and units are accepted too. In the Final Task, learners should state units.

Why include 'believable' cards that are correct?
So learners judge each answer on its merits, rather than assuming every card hides a mistake.

What about the end of the unit?
This is the last lesson. A short end-of-unit assessment follows; its approach is still to be agreed.

How does this connect to earlier lessons?
It uses substitution (Lesson 2), forming formulae (Lesson 3) and rearranging (Lesson 5) together, with checking as the new habit.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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